

SYSC 5108 Exam Study: Assignment 2, Question 5
Logistic regression gradient for 0/1 and $-1/+1$ label encodings
Source question: SYSC 5108 W26 Assignment 2.pdf, Question 5

Question

Consider training a logistic regression model with $K = 2$ classes. First assume $y \in \{0, 1\}$ and

$$J(w) = -y \log(\sigma(z)) - (1 - y) \log(1 - \sigma(z)), \quad z = w^T x, \quad x_0 = 1.$$

Show that

$$\nabla_w J(w) = -x(y - \sigma(z)).$$

Then change the encoding to $y \in \{-1, 1\}$, show the result again, and explain the impact.

Step 1: Recall the sigmoid derivative

Chapter 3 slides 32–34 define the logistic sigmoid:

$$\sigma(z) = \frac{1}{1 + e^{-z}}.$$

The derivative that we need is

$$\frac{d\sigma(z)}{dz} = \sigma(z)(1 - \sigma(z)).$$

Also, since

$$z = w^T x,$$

the gradient of z with respect to w is

$$\nabla_w z = x.$$

Part (a): Derivation for $y \in \{0, 1\}$

Let

$$s = \sigma(z).$$

The loss is

$$J(w) = -y \log(s) - (1 - y) \log(1 - s).$$

First differentiate with respect to s :

$$\frac{dJ}{ds} = -\frac{y}{s} + \frac{1 - y}{1 - s}.$$

Now use

$$\frac{ds}{dz} = s(1 - s).$$

By the chain rule,

$$\frac{dJ}{dz} = \frac{dJ}{ds} \frac{ds}{dz} = \left(-\frac{y}{s} + \frac{1 - y}{1 - s} \right) s(1 - s).$$

Simplify:

$$\frac{dJ}{dz} = -y(1-s) + (1-y)s.$$

Expand the terms:

$$\frac{dJ}{dz} = -y + ys + s - ys = s - y.$$

Since $s = \sigma(z)$,

$$\frac{dJ}{dz} = \sigma(z) - y.$$

Finally apply the chain rule through $z = w^T x$:

$$\nabla_w J(w) = \frac{dJ}{dz} \nabla_w z = (\sigma(z) - y)x.$$

This is the same as

$$\boxed{\nabla_w J(w) = -x(y - \sigma(z))}.$$

Step 2: Interpret the 0/1 Gradient

For $y = 1$:

$$\nabla_w J(w) = -(1 - \sigma(z))x.$$

Gradient descent uses

$$w_{\text{new}} = w - \epsilon \nabla_w J(w),$$

so the update adds a positive multiple of x when the positive class is underpredicted.

For $y = 0$:

$$\nabla_w J(w) = \sigma(z)x.$$

Gradient descent subtracts a positive multiple of x , pushing the logit $z = w^T x$ downward.

This matches the intuition: if the target is 1, push the probability upward; if the target is 0, push the probability downward.

Part (b): Change the Encoding to $y \in \{-1, 1\}$

When labels are encoded as -1 and $+1$, do not plug $y = -1$ directly into the 0/1 cross-entropy formula. That formula assumes y is an indicator variable. Instead, use the equivalent binary logistic loss written in margin form:

$$J(w) = \log(1 + e^{-yz}), \quad y \in \{-1, 1\}.$$

Here,

$$yz$$

is the signed margin. If yz is large and positive, the example is classified correctly with high confidence. If yz is negative, the example is on the wrong side of the decision boundary.

Differentiate with respect to z :

$$\frac{dJ}{dz} = \frac{1}{1 + e^{-yz}} \cdot e^{-yz}(-y).$$

So

$$\frac{dJ}{dz} = -\frac{ye^{-yz}}{1 + e^{-yz}}.$$

Divide numerator and denominator by e^{-yz} :

$$\frac{dJ}{dz} = -\frac{y}{1 + e^{yz}}.$$

Using

$$\sigma(-yz) = \frac{1}{1 + e^{yz}},$$

we get

$$\frac{dJ}{dz} = -y\sigma(-yz).$$

Now use $\nabla_w z = x$:

$$\boxed{\nabla_w J(w) = -y x \sigma(-yz)}.$$

Equivalently,

$$\boxed{\nabla_w J(w) = -\frac{yx}{1 + e^{yz}}}.$$

Step 3: Show Consistency with the 0/1 Formula

Let the old label be $t \in \{0, 1\}$ and the new label be

$$y = 2t - 1.$$

If $t = 1$, then $y = 1$. The $-1/+1$ gradient becomes

$$-yx\sigma(-yz) = -x\sigma(-z).$$

Since

$$\sigma(-z) = 1 - \sigma(z),$$

this is

$$-x(1 - \sigma(z)),$$

which matches the 0/1 result for $t = 1$:

$$x(\sigma(z) - 1).$$

If $t = 0$, then $y = -1$. The $-1/+1$ gradient becomes

$$-(-1)x\sigma(z) = x\sigma(z),$$

which matches the 0/1 result for $t = 0$:

$$x(\sigma(z) - 0) = x\sigma(z).$$

So the two encodings describe the same binary logistic regression objective if the loss is written correctly for each encoding.

Step 4: Explain the Impact of Changing the Encoding

Changing from 0/1 labels to $-1/+1$ labels does not change the decision boundary. In both cases, the boundary is

$$z = w^T x = 0.$$

For the 0/1 encoding,

$$p(y = 1 \mid x) = \sigma(z),$$

and the usual threshold is $p \geq 0.5$, which is equivalent to $z \geq 0$.

For the $-1/+1$ encoding, the loss is written using the margin yz . This form is often cleaner because:

- Correct and confident examples have large positive yz and therefore a small loss.
- Incorrect examples have negative yz and therefore a large loss.
- The gradient magnitude is controlled by $\sigma(-yz)$, so examples that are badly misclassified receive a larger update.

The main caution is that the old 0/1 cross-entropy formula cannot be reused unchanged with $y = -1$. You must rewrite it as

$$J(w) = \log(1 + e^{-yz}).$$

Concise Exam Answer

For $y \in \{0, 1\}$, let $s = \sigma(z)$. Then

$$J(w) = -y \log(s) - (1 - y) \log(1 - s).$$

The chain rule gives

$$\frac{dJ}{dz} = \left(-\frac{y}{s} + \frac{1-y}{1-s} \right) s(1-s) = s - y = \sigma(z) - y.$$

Since $\nabla_w z = x$,

$$\nabla_w J(w) = (\sigma(z) - y)x = -x(y - \sigma(z)).$$

For $y \in \{-1, 1\}$, use

$$J(w) = \log(1 + e^{-yz}).$$

Then

$$\frac{dJ}{dz} = -\frac{y}{1 + e^{yz}} = -y\sigma(-yz),$$

so

$$\nabla_w J(w) = -yx\sigma(-yz) = -\frac{yx}{1 + e^{yz}}.$$

The impact is mostly notational and algebraic: $-1/+1$ labels express the loss in terms of the signed margin yz , while the 0/1 form expresses it as Bernoulli cross-entropy. The decision boundary remains $w^T x = 0$ when the loss is written consistently.

Cross References Used

- Assignment: SYSC 5108 W26 Assignment 2.pdf, Question 5.
- Local submitted Assignment 2 PDF checked for comparison, Question 5 response.
- Slides: Chapter 3 slide deck, slides 32–34 on the logistic sigmoid and binary logistic regression.
- Slides: Chapter 3 slide deck, logistic regression training slide on log-likelihood and negative log-likelihood.
- Textbook: Goodfellow, Bengio, and Courville, *Deep Learning*, Chapter 3, Section 3.10 on the logistic sigmoid.
- Textbook: Goodfellow, Bengio, and Courville, *Deep Learning*, Chapter 6, Section 6.2.2.2 on sigmoid output units for Bernoulli output distributions and maximum likelihood.